Practice Quiz: Lecture 6 (Part 2)

Topic: Synchronous & Statistical TDM, Interleaving, Bit Padding **Questions:** 12 | **Total Marks:** 25

> **Instructions:** Answer all 12 questions. Calculate values precisely and clearly write values with appropriate units (μs, ms, kbps, etc.). Show step-by-step working.

1. Matching Multiplexing Paradigms

Match each description with the most suitable multiplexing technology (**FDM**, **TDM**, or **WDM**):

1. Multiplexing technology designed specifically for optical fiber propagation using light wavelengths.	Answer: ----------
2. Digital multiplexing scheme categorized into synchronous and asynchronous (statistical) modes.	Answer: ----------
3. Traditional analog multiplexing technique used for terrestrial TV broadcast and dial-up FSK modems.	Answer: ----------
4. Divides link transmission capacity strictly by time slices rather than frequency bands.	Answer: ----------

2. (Calculation) Multiplexing Identical Low-Speed Connections

Six 2-kbps connections are multiplexed together using Synchronous TDM. The unit of interleaving is 1 bit.

1. What is the bit duration of an input source channel (in ms)?	Answer: ----------
2. What is the transmission bit rate of the aggregated link (in kbps)?	Answer: ----------
3. What is the bit duration on the output aggregated link (in ms)?	Answer: ----------

3. (Calculation) Synchronous TDM Timing Analysis

A synchronous TDM multiplexer combines four 50-kbps digital channels using a time slot size of 2 bits per input channel per frame. Calculate each parameter:

1. Frame rate (in kilo-frames/s):	Answer: ----------
2. Frame duration (in millisecond):	Answer: ----------
3. Frame duration (in μs):	Answer: ----------
4. Bit rate of the composite link (in kbps):	Answer: ----------
5. Bit duration / bit interval on the composite link (in μs):	Answer: ----------

4. (Calculation) Byte-Interleaved Synchronous TDM

A synchronous TDM multiplexes five 64-kbps digital voice channels. Interleaving is performed byte-by-byte ($1\ \text{byte} = 8\ \text{bits}$ per slot). Calculate:

a. The number of bits in each output frame.
b. The output frame rate (in frames/s).
c. The duration of one time slot on the output link (in μs).

5. Dynamic Allocation & Multiplexing Characteristics

In ---------- , time slots are dynamically allocated on demand rather than fixed permanently, significantly improving bandwidth efficiency across bursty digital transmission lines.

- □ a. Synchronous TDM
- □ b. Statistical (Asynchronous) TDM
- □ c. Frequency Division Multiplexing (FDM)
- □ d. Wavelength Division Multiplexing (WDM)

6. Which of the following multiplexing schemes is NOT associated with traditional physical circuit switching?

- □ a. Frequency Division Multiplexing (FDM)
- □ b. Wavelength Division Multiplexing (WDM)
- □ c. Synchronous Time Division Multiplexing (STDM)
- □ d. Asynchronous / Statistical Time Division Multiplexing (ATDM)

7. (Comprehensive Problem) Bit Padding / Pulse Stuffing Design

Three digital data sources with transmission rates of $100\ \text{kbps}$, $300\ \text{kbps}$, and $125\ \text{kbps}$ are combined onto a single multiplexed channel using Synchronous TDM. An integer slot allocation ratio based on the lowest common base rate is required.

1. Which input channel requires bit padding?	Answer: ----------
2. What is the required bit padding rate for that channel (in kbps)?	Answer: ----------
3. Assuming $1\ \text{slot} = 1\ \text{bit}$, how many total bits are transmitted per frame?	Answer: ----------
4. What is the composite frame rate (in kilo-frames/s)?	Answer: ----------
5. What is the resulting frame duration (in μs)?	Answer: ----------
6. What is the aggregated output transmission bit rate (in kbps)?	Answer: ----------

8. (Calculation) Pulse Stuffing with Different Channel Rates

Four digital channels with bit rates of $200\ \text{kbps}$, $200\ \text{kbps}$, $400\ \text{kbps}$, and $150\ \text{kbps}$ are multiplexed via Synchronous TDM. The frame rate is set based on a base channel rate of $200\ \text{kbps}$.

a. What padding rate must be added to the 150-kbps stream?
b. Assuming $1\ \text{bit}$ per slot for the 200-kbps streams, how many slots does each frame carry?
c. What is the total bit rate of the aggregated link?

9. (Calculation) Framing Bit Overhead

Four 1-Mbps connections are multiplexed using Synchronous TDM with $1\ \text{bit}$ interleaved from each channel. To ensure receiver synchronization, **one framing bit** is added to every output frame.

a. What is the size of each output frame in bits?
b. What is the output frame rate (in frames/s)?
c. What is the total data transmission rate of the output link (in Mbps)?

10. Why do slots in Statistical (Asynchronous) TDM require addressing overhead, whereas slots in Synchronous TDM do not?

- □ a. Synchronous TDM transmits analog signals, so channel addressing is handled by bandpass filters.
- □ b. In Synchronous TDM, the identity of the source is implicitly determined by the slot's predefined position within the frame; in Statistical TDM, slots are dynamically assigned only to active sources.
- □ c. Statistical TDM does not use time division, making addresses mandatory for carrier demodulation.
- □ d. Synchronous TDM frames contain error correction headers that substitute for destination addresses.

11. (Conceptual) Idle Slot Waste in Synchronous TDM

Which statement best characterizes the behavior and inefficiency of Synchronous TDM when an input source has no active data to transmit?

- □ a. The multiplexer collapses the frame and transmits only data from active sources.
- □ b. The slot assigned to that source is omitted, and the frame duration shrinks dynamically.
- □ c. The dedicated slot for that source remains reserved and is transmitted empty (idle), wasting link capacity.
- □ d. The multiplexer buffers data from the next channel into the vacant slot.

12. (Calculation) Statistical Multiplexing Capacity Comparison

Ten terminals share a synchronous communication link. Each terminal transmits at $9600\ \text{bps}$ when active, but is active only 10% of the time.

a. What link capacity is strictly required if using **Synchronous TDM**?
b. What is the **average aggregated throughput** generated by all terminals combined?
c. Why is Statistical TDM advantageous in this scenario?

Answer Key & Detailed Explanations: Lecture 6 (Part 2)

Question 1: Matching

Answers:

1. **WDM** (Wavelength Division Multiplexing)
2. **TDM** (Time Division Multiplexing)
3. **FDM** (Frequency Division Multiplexing)
4. **TDM** (Time Division Multiplexing)

Question 2: Timing Analysis of 6 Connections

Explanation:

1. Input bit duration:

$$T_b = \frac{1}{\text{Bit rate}} = \frac{1}{2000 \text{ bps}} = 0.0005 \text{ s} = 0.5 \text{ ms}$$

2. Aggregated output rate:

$$\text{Link Rate} = 6 \times 2 \text{ kbps} = 12 \text{ kbps}$$

3. Output bit duration:

$$T_{\text{out}} = \frac{1}{12000 \text{ bps}} = \frac{0.5 \text{ ms}}{6} \approx 0.0833 \text{ ms} = 83.33 \ \mu\text{s}$$

Question 3: Synchronous TDM Analysis

Explanation:

1. Frame rate: Each channel delivers 50 kbps with 2 bits per slot per frame:

$$\text{Frame Rate} = \frac{50 \text{ kbps}}{2 \text{ bits/frame}} = 25 \text{ k-frames/s}$$

2. Frame duration (ms):

$$T_f = \frac{1}{25000 \text{ frames/s}} = 0.00004 \text{ s} = 0.04 \text{ ms}$$

3. Frame duration (μs):

$$0.04 \text{ ms} \times 1000 = 40 \ \mu\text{s}$$

4. Aggregated bit rate:

$$\text{Bits per frame} = 4 \text{ channels} \times 2 \text{ bits/slot} = 8 \text{ bits}$$

$$\text{Link Bit Rate} = 25 \text{ k-frames/s} \times 8 \text{ bits/frame} = 200 \text{ kbps}$$

$$(\text{Alternatively: } 4 \times 50 \text{ kbps} = 200 \text{ kbps})$$

5. Bit interval on link:

$$T_{\text{link bit}} = \frac{1}{200 \text{ kbps}} = 0.000005 \text{ s} = 0.005 \text{ ms} = 5 \ \mu\text{s}$$

Question 4: Byte-Interleaved Synchronous TDM

Explanation:

a. Bits per frame:

$$\text{Bits/frame} = 5 \text{ channels} \times 8 \text{ bits/channel} = 40 \text{ bits}$$

b. Frame rate:

$$\text{Frame Rate} = \frac{64 \text{ kbps}}{8 \text{ bits/slot}} = 8000 \text{ frames/s}$$

c. Slot duration:

$$T_{\text{frame}} = \frac{1}{8000 \text{ frames/s}} = 125 \ \mu\text{s}$$

$$T_{\text{slot}} = \frac{T_{\text{frame}}}{5 \text{ slots}} = \frac{125 \ \mu\text{s}}{5} = 25 \ \mu\text{s}$$

Question 5: b (Statistical / Asynchronous TDM)

Explanation: Statistical TDM dynamically allocates slots to lines that currently have data to send. Idle lines are skipped, eliminating wasted slots and yielding greater bandwidth utilization.

Question 6: d (Asynchronous Time Division Multiplexing)

Explanation: FDM, WDM, and Synchronous TDM establish predetermined, dedicated physical channels or time slots, making them standard physical foundations for **circuit switching**. In contrast, Asynchronous/Statistical TDM dynamically allocates resources with headers, forming the baseline for **packet switching**.

Question 7: Bit Padding Problem

Explanation:

1. The base rate is 100 kbps. Channel 2 (300 kbps) is an integer multiple: $300/100 = 3$. Channel 3 (125 kbps) cannot be divided evenly: $125/100 = 1.25$. Thus, **Channel 3 (125 kbps)** requires bit padding.
2. Padding rate: It must be padded up to the next integer multiple of 100 kbps, which is 200 kbps:

$$\text{Padding Rate} = 200 \text{ kbps} - 125 \text{ kbps} = 75 \text{ kbps}$$

3. Slots per frame:

$$\text{Channel 1 (100 kbps)} \implies 1 \text{ slot}$$

$$\text{Channel 2 (300 kbps)} \implies 3 \text{ slots}$$

$$\text{Channel 3 (padded to 200 kbps)} \implies 2 \text{ slots}$$

$$\text{Total Slots} = 1 + 3 + 2 = 6 \text{ slots (or bits, assuming 1 bit/slot)}$$

4. Frame rate: Dictated by the base rate:

$$\text{Frame Rate} = 100 \text{ kilo-frames/second}$$

5. Frame duration:

$$T_f = \frac{1}{100000 \text{ frames/s}} = 0.00001 \text{ s} = 10 \ \mu\text{s}$$

6. Output bit rate:

$$\text{Aggregate Bit Rate} = 100 \text{ k-frames/s} \times 6 \text{ bits/frame} = 600 \text{ kbps}$$

Question 8: Pulse Stuffing with Different Channel Rates

Explanation:

a. The 150-kbps channel is padded up to the base rate of 200 kbps:

$$\text{Padding Rate} = 200 \text{ kbps} - 150 \text{ kbps} = 50 \text{ kbps}$$

b. Slots per frame:

$$\text{Ch 1 (200k)} \rightarrow 1 \text{ slot}, \quad \text{Ch 2 (200k)} \rightarrow 1 \text{ slot}, \quad \text{Ch 3 (400k)} \rightarrow 2 \text{ slots}, \quad \text{Ch 4 (200k padded)} \rightarrow 1 \text{ slot}$$

$$\text{Total Slots} = 1 + 1 + 2 + 1 = 5 \text{ slots/frame}$$

c. Link bit rate:

$$\text{Frame Rate} = 200 \text{ k-frames/s}$$

$$\text{Link Bit Rate} = 200 \text{ k-frames/s} \times 5 \text{ bits/frame} = 1000 \text{ kbps} = 1 \text{ Mbps}$$

Question 9: Framing Bit Overhead

Explanation:

a. Frame size:

$$\text{Frame Size} = (4 \text{ channels} \times 1 \text{ bit}) + 1 \text{ framing bit} = 5 \text{ bits}$$

b. Frame rate:

$$\text{Frame Rate} = \frac{1 \text{ Mbps}}{1 \text{ bit/channel}} = 1{,}000{,}000 \text{ frames/s}$$

c. Output link bit rate:

$$\text{Link Bit Rate} = 1{,}000{,}000 \text{ frames/s} \times 5 \text{ bits/frame} = 5 \text{ Mbps}$$

Question 10: b

Explanation: In Synchronous TDM, time slots are assigned in strict, cyclical, round-robin order. The receiver knows exactly which transmitter owns a slot by counting its position inside the frame. In Statistical TDM, slots are assigned on the fly only to lines with active data; each slot must therefore carry addressing metadata so the demultiplexer knows which destination line to route the payload to.

Question 11: c

Explanation: In Synchronous TDM, time slots are permanently dedicated to specific channels regardless of whether they have data to send. If a station is silent, its allocated time slot passes through the link empty, which can result in low channel efficiency for bursty traffic.

Question 12: Statistical Multiplexing Analysis

Explanation:

a. Synchronous TDM requires static reservation for peak capacity of all channels:

$$\text{Required Link Capacity} = 10 \times 9600 \text{ bps} = 96 \text{ kbps}$$

b. Average generated throughput across all terminals:

$$\text{Average Throughput} = 10 \times (9600 \text{ bps} \times 0.10) = 9.6 \text{ kbps}$$

c. In this scenario, Synchronous TDM wastes $\approx 90\%$ of link capacity transmitting idle slots. Statistical TDM can support all 10 devices using a significantly lower link rate (e.g., 19.2 kbps or 28.8 kbps including addressing overhead), providing large cost and bandwidth savings.